MATH329 · OPERATIONS RESEARCH

Lecture 3 Standard Form

Midterm summary · Chapter 3 · the algebraic solution

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01 Why do we need the standard form?

The graph works for two variables only. With more variables we solve algebraically with the **Simplex** method, which moves from one corner to a better corner until it reaches the optimum.

> The Simplex Method attempts to move from one corner point of the solution space to a better corner point until the optimum is found.

Before simplex starts, the problem must be written in the **Standard Form** (also called **Equation form**). This is a fixed question in every midterm, worth 5 to 8 marks.

Appeared in the midterm (461)

The question: what does standard form mean? The expected answer, word for word:

> 1. All decision variables are **nonnegative**.
> 2. All constraints (with the exception of the non-negativity of the variables) are **equations with nonnegative RHS**.

02 The conversion rules

Case	What to do	Example
A $\leq$ constraint	Add a **slack variable** with a plus sign.	$x_1 + 3x_2 \leq 6$ $x_1 + 3x_2 + s_1 = 6$
A $\geq$ constraint	Subtract a **surplus variable**, with a minus sign.	$3x_1 + 2x_2 \geq 6$ $3x_1 + 2x_2 - s_2 = 6$
An $=$ constraint	It stays as it is, with no new variable.	$x_1 + x_2 = 5$
A negative right-hand side	Multiply the whole constraint by -1. If it is an inequality, its sign flips.	$3x_1 + 2x_2 \geq -2$ $-3x_1 - 2x_2 \leq 2$
A variable **unrestricted in sign**	Replace it everywhere, including the objective, by the difference of two nonnegative variables.	$x_2 = x_2^+ - x_2^-$
The objective function	Add the new variables with zero coefficients.	$z = 2x_1 + 3x_2 + 0s_1 + 0s_2$
The last line	All variables, old and new, are ≥ 0.	$x_1, x_2, s_1, s_2 \geq 0$

The order matters

First fix the negative right-hand side, then substitute the unrestricted variable, and add the slack and surplus variables last. Adding a slack before multiplying by -1 gives it the wrong sign.

03 The slide examples

1 SLACK Example 3.1

Maximize $z = 2x_1 + 3x_2$
subject to $x_1 + 3x_2 \le 6$
$3x_1 + 2x_2 \le 6$
$x_1, x_2 \ge 0$

Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$
subject to $x_1 + 3x_2 + s_1 = 6$
$3x_1 + 2x_2 + s_2 = 6$
$x_1, x_2, s_1, s_2 \ge 0$

2 SURPLUS The second constraint is now ≥

Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$	
subject to $x_1 + 3x_2 + s_1 = 6$	s_1 slack
$3x_1 + 2x_2 - s_2 = 6$	s_2 surplus
$x_1, x_2, s_1, s_2 \ge 0$	

3 NEGATIVE RHS The second constraint is $3x_1 + 2x_2 \geq -2$

$$3x_1 + 2x_2 \geq -2 \quad \times (-1) \quad \rightarrow \quad -3x_1 - 2x_2 \leq 2$$

$$\begin{aligned} &\text{Maximize } z = 2x_1 + 3x_2 + 0s_1 + 0s_2 \\ &\text{subject to} \quad x_1 + 3x_2 + s_1 = 6 \\ &\qquad\qquad -3x_1 - 2x_2 + s_2 = 2 \\ &\qquad\qquad x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

4 UNRESTRICTED x_1 is unrestricted

$$\begin{aligned} &\text{Maximize } z = x_1 + 3x_2 \\ &\text{subject to} \quad x_1 + x_2 \leq 2 \\ &\qquad\qquad -x_1 + x_2 \leq 4 \\ &\qquad\qquad x_1 \text{ unrestricted}, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} &\text{Maximize } z = x_1^+ - x_1^- + 3x_2 + 0s_1 + 0s_2 \\ &\text{subject to} \quad x_1^+ - x_1^- + x_2 + s_1 = 2 \\ &\qquad\qquad -x_1^+ + x_1^- + x_2 + s_2 = 4 \\ &\qquad\qquad x_1^+, x_1^-, x_2, s_1, s_2 \geq 0 \end{aligned}$$

04 The midterm question: every rule in one problem

This question appeared in the same shape in 461 and 462 with different numbers. This is

the 462 version (8 marks):

Maximum $Z = 8x_1 - 2x_2$

S.to $2x_1 + x_2 \leq 12 \cdot x_1 - 5x_2 \geq 6 \cdot 4x_2 = -7$

$x_1 \geq 0$, x_2 unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). "Don't solve the problem"

1 The methods

`Graphical method and Algebraic method.`

The graphical method applies because there are two variables, and the algebraic method always applies.

2 The negative right-hand side

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

3 Substitute $x_2 = x_2^+ - x_2^-$ everywhere

$-2x_2 \quad \rightarrow \quad -2x_2^+ + 2x_2^-$

$x_2 \quad \rightarrow \quad x_2^+ - x_2^-$

$-5x_2 \quad \rightarrow \quad -5x_2^+ + 5x_2^-$

$-4x_2 \quad \rightarrow \quad -4x_2^+ + 4x_2^-$

4

Add s_1 to the $\leq$ constraint and subtract s_2 from the $\geq$ constraint

Maximum	$Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$
S.to	$2x_1 + x_2^+ - x_2^- + s_1 = 12$
	$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$
	$-4x_2^+ + 4x_2^- = 7$
	$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

In the answer key: two marks for the methods, two for the objective function, and one for each of the other four lines.

05 Basic solutions: from the graph to algebra

After the standard form we have n variables and m equations, with more variables than equations. To solve:

1. Choose n − m variables and set them equal to zero (nonbasic)
2. Solve the m equations for the remaining m variables (basic)
3. The result is a basic solution
4. If all its values are ≥ 0 it is a Basic Feasible Solution (BFS)

Term	Meaning
Nonbasic variables	The `n − m` variables that we set to zero.
Basic variables	The `m` variables that we solve for.
Basic solution	The unique solution of the system after setting the nonbasic variables to zero.
Basic Feasible Solution	A basic solution with no negative values. Every BFS corresponds to a corner of the feasible region.

The maximum number of basic solutions

$$C(n, m) = \frac{n!}{m!\,(n-m)!}$$

$n = 4, m = 2$: $\quad C(4, 2) = \frac{4!}{2! \cdot 2!} = \frac{24}{4} = 6$

Appeared in the midterm (471)

True/false items, both true:

> A basic feasible solution is a feasible solution to LPP which is also the basic solution. (T)
> A basic feasible solution is called degenerate if value of at least one basic variable is zero. (T)

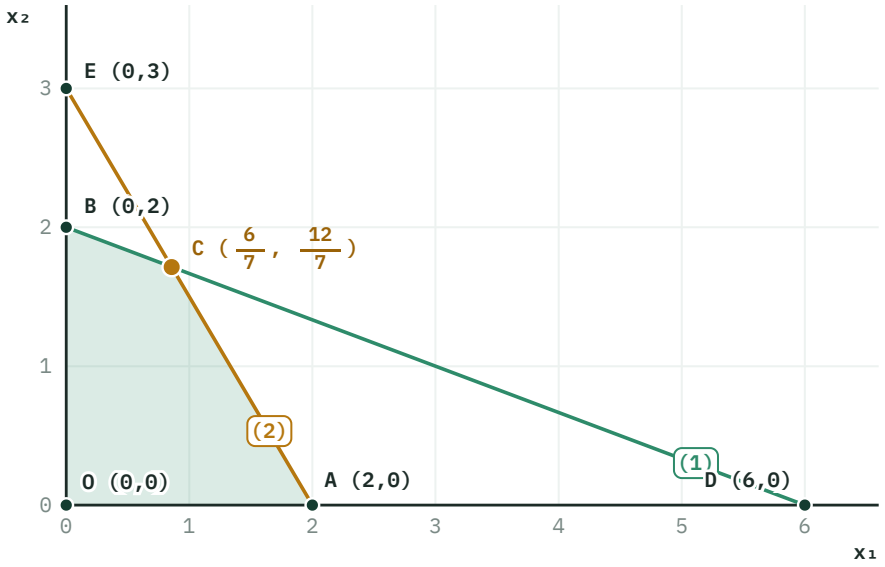
06 Example 3.2: all the basic solutions

Maximize $z = 2x_1 + 3x_2 + 0s_1 + 0s_2$
subject to $x_1 + 3x_2 + s_1 = 6$
$3x_1 + 2x_2 + s_2 = 6$
$x_1, x_2, s_1, s_2 \geq 0$

Here $n = 4$ and $m = 2$, so we set two variables to zero each time and solve for the rest. There are 6 cases:

Nonbasic (= 0)	Basic	Basic solution	Feasible?	z	Corner
s_1, s_2	x_1, x_2	$(\frac{6}{7}, \frac{12}{7}, 0, 0)$	Yes	$\frac{48}{7}$	C
x_2, s_2	x_1, s_1	$(2, 0, 4, 0)$	Yes	4	A
x_2, s_1	x_1, s_2	$(6, 0, 0, -12)$	No	—	D
x_1, s_2	x_2, s_1	$(0, 3, -3, 0)$	No	—	E
x_1, s_1	x_2, s_2	$(0, 2, 0, 2)$	Yes	6	B
x_1, x_2	s_1, s_2	$(0, 0, 6, 6)$	Yes	0	O

The solution is written in the order (x_1, x_2, s_1, s_2). One row as an example: set x_1 and s_1 to zero, so the first equation becomes $3x_2 = 6$ giving $x_2 = 2$, and the second becomes $2(2) + s_2 = 6$ giving $s_2 = 2$.



Optimum $z = \frac{48}{7}$ **at** $x_1 = \frac{6}{7}$, $x_2 = \frac{12}{7}$ **(corner C)**

The two infeasible solutions, D and E, are on the graph but outside the shaded region. The four feasible ones are exactly the corners of the region.

07 Example 3.3: with an unrestricted variable

$$\text{Maximize } z = x_1^+ - x_1^- + 3x_2 + 0s_1 + 0s_2$$

$$\text{subject to} \quad x_1^+ - x_1^- + x_2 + s_1 = 2$$

$$-x_1^+ + x_1^- + x_2 + s_2 = 4$$

$$x_1^+,\ x_1^-,\ x_2,\ s_1,\ s_2 \geq 0$$

Here `n = 5` and `m = 2`, so the maximum is `C(5, 2) = 10`. One case has no solution, so the actual number is 9. The number of basic solutions can be less than the maximum.

Nonbasic (= 0)	Basic	Basic solution	Feasible?	z
x_2, s_1, s_2	x_1^+, x_1^-	no solution	–	–
x_1^-, s_1, s_2	x_1^+, x_2	(−1, 0, 3, 0, 0)	No	–
x_1^-, x_2, s_2	x_1^+, s_1	(−4, 0, 0, 6, 0)	No	–
x_1^-, x_2, s_1	x_1^+, s_2	(2, 0, 0, 0, 6)	Yes	2
x_1^+, s_1, s_2	x_1^-, x_2	(0, 1, 3, 0, 0)	Yes	**8**
x_1^+, x_2, s_2	x_1^-, s_1	(0, 4, 0, 6, 0)	Yes	−4
x_1^+, x_2, s_1	x_1^-, s_2	(0, −2, 0, 0, 6)	No	–
x_1^+, x_1^-, s_2	x_2, s_1	(0, 0, 4, −2, 0)	No	–
x_1^+, x_1^-, s_1	x_2, s_2	(0, 0, 2, 0, 2)	Yes	6
x_1^+, x_1^-, x_2	s_1, s_2	(0, 0, 0, 2, 4)	Yes	0

Optimum $z = 8$ at $x_1^- = 1$, $x_2 = 3$ so $x_1 = 0 - 1 = -1$, $x_2 = 3$

The slide's table marks the solution `(0, −2, 0, 0, 6)` as feasible, but $x_1^- = -2$ is negative, so it is not feasible. The table here is corrected.

08 Example 3.5: absolute values

Convert the following optimization problem into a LPP:

Maximum $z = \max\{ |2x_1 + 3x_2|, |3x_1 - 7x_2| \}$

Subject to: $x_1 + x_2 \le 2$, $-x_1 + x_2 \le 4$, $x_1, x_2 \ge 0$

The idea: call the larger value `y`. Since `|a| ≤ y` means `−y ≤ a ≤ y`, each absolute value gives two linear constraints:

Maximize $Z = y$	
subject to	$x_1 + x_2 \leq 2$
	$-x_1 + x_2 \leq 4$
	$2x_1 + 3x_2 - y \leq 0$
	$-2x_1 - 3x_2 - y \leq 0$
	$3x_1 - 7x_2 - y \leq 0$
	$-3x_1 + 7x_2 - y \leq 0$
	$x_1, x_2, y \geq 0$

This is the slide's form. Note: when maximizing y with only constraints of the form $y \geq \ldots$, y can grow without limit. This conversion is valid when the problem is a minimization.

09 Common mistakes

Mistake	Correct
Adding a slack to a $\geq$ constraint	A surplus is subtracted from a $\geq$ constraint: $-s_2$.
Adding a variable to an $=$ constraint	It gets no slack and no surplus.
Leaving a negative right-hand side	$4x_2 = -7$ becomes $-4x_2 = 7$.
Substituting the unrestricted variable in the constraints but not in the objective	It is substituted everywhere it appears.
Losing signs with a negative coefficient	$-5x_2$ becomes $-5x_2^+ + 5x_2^-$.
Forgetting the last line	$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

10 Test yourself

1. What does the standard form mean? (two requirements)
2. What is the difference between a slack and a surplus variable?
3. What is the maximum number of basic solutions when $n = 5$ and $m = 3$?
4. Write in standard form: $\text{Min } Z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 9$, $2x_1 - x_2 \geq 4$, $3x_2 = -6$, with $x_1 \geq 0$ and x_2 unrestricted.

Answers

#	Answer
1	All decision variables are nonnegative, and all constraints are equations with nonnegative RHS.
2	A slack is added to a $\leq$ constraint, and a surplus is subtracted from a $\geq$ constraint.
3	$$C(5, 3) = \frac{5!}{3! \cdot 2!} = 10$$
4	$\text{Min } Z = 3x_1 + 2x_2^+ - 2x_2^- + 0s_1 + 0s_2$ $x_1 + x_2^+ - x_2^- + s_1 = 9$ $2x_1 - x_2^+ + x_2^- - s_2 = 4$ $-3x_2^+ + 3x_2^- = 6$ $x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$